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1. Morse theory and geodesics I — Zhen Gao

November 2, 2022

Abstract. In this first talk, I will review quickly the basic Morse theory, and relate it to the study of (closed) geodesics on Riemannian manifolds.

2. Hamiltonian torus actions and the convexity theorem — Sungho Kim

November 9, 2022

Abstract. In this talk, I will explain about Hamiltonian torus actions on symplectic manifolds. After investigating examples, I will give a sketch of proof for the famous convexity theorem, proven by Atiyah and Guillemin-Sternberg. The key ingredient is Morse-Bott theory which is a generalization of Morse theory.

3. Morse theory and geodesics II — Zhen Gao

November 16, 2022

Abstract. Continuing from the first talk, more on geodesics and Riemannian geometry will be discussed this week. I will expose how Morse theory can be applied to study the topology of loop spaces via minimal geodesics, which is basically due to the seminal work by Raoul Bott from the last century, leading to the famous Bott's periodicity.

4. An Introduction to Topological K-theory I — Andreas Hube & Helge Frerichs

November 23, 2022

Abstract. The upcoming three talks will give a basic introduction to topological K-theory. The first talk will deal with the classification problem for vector bundles and its relation to (unstable) homotopy theory. The main goal then is to lead the way towards the definition of K-theory.

5. An Introduction to Topological K-theory II — Andreas Hube & Helge Frerichs

November 30, 2022

Abstract. We will talk about the Product Theorem, Bott periodicity and K-Theory as a cohomology theory.

6. An Introduction to Topological K-theory III — Andreas Hube & Helge Frerichs

December 7, 2022

Abstract. As an application for K-theory, we will discuss the initial problem of parallelizability of spheres. In fact, we show that S^n is parallelizable iff. $n = 1, 3$ or 7.

7. An introduction to C*-algebra theory and the classification of AF-algebras — Airi Takeuchi

December 14, 2022

Abstract. Operator theoretic K-theory is a powerful tool for understanding the structure of C*-algebras. In this talk, we will first study the basic properties of C*-algebras and define K_0 groups from the equivalence classes of projections. Finally, we introduce Elliott's theorem that the easiest class of non-commutative infinite-dimensional C*-algebras, AF-algebras, can be completely classified by their K_0 groups.

8. An Introduction to Topological K-theory IV — Andreas Hube & Helge Frerichs

December 21, 2022

Abstract. In this talk we will conclude the proof that division algebras only exist in dimensions 1,2,4 and 8. To do this we will introduce Adams operations and show how this helps to reduce the question of the existence of a division algebra to an elementary number theoretic problem.

9. Low dimensional topology I: Mapping Class Group — Zhen Gao

January 11, 2023

Abstract. We will learn what a mapping class group for an oriented smooth surface (possibly with boundaries) is, and study a few examples explicitly. The main goal is to show the generation result on mapping class group by Dehn twists.

10. Variational Treatment of the Schrödinger Equation — Martin Konrad

January 25, 2023

Abstract. Using the calculus of variations, we prove the existence of a ground state solution to the Schrödinger equation for the "right" kind of potential.

11. Low dimensional topology II: Dehn twists — Zhen Gao

February 1, 2023

Abstract. Continuing from my last talk, we show how the mapping class group of torus is explicitly generated by Dehn twists of two basic simple closed curves. We then give proof of the generation result on general mapping class group for oriented surfaces.

12. Low dimensional topology III: Dehn surgery — Zhen Gao

February 8, 2023

Abstract. After an briefly overviw on the handle body theory and Heegaard splitting of 3-manifolds, we will learn another technique by Dehn on surgery. Eventually, the combination of Dehn's operations leads to the well-known result that the oriented 3-cobordism group is trivial.